M.Tech (CSE) - I Year

CS655 - Drone Technologies (July 2026)

Assignment **Marks: 20**

Q.No.	Team membe	Team mem	Team memb	Problem statement (Use Drone to solve)
1	206126013	206126009	206126032	Simulate a drone equipped with a camera to capture crop images. Integrate an image classification or object detection model to identify potentially diseased plants and mark their locations on a field map.
2	206126019	206126022	206126017	Simulate a drone patrolling a forest environment to identify fire locations using simulated thermal-camera data.
3	206126002	206126003	206126027	Develop a drone simulation that surveys a virtual wildlife reserve, detects animals using camera data, and estimates the number of animals observed without repeatedly counting the same animal.
4	206126005	206126031	206126023	Simulate a drone monitoring a road intersection. Use computer vision to count vehicles and classify traffic density as low, medium, or high.
5	206126016	206126001	206126012	Simulate a delivery drone transporting a package between specified locations while avoiding obstacles and respecting predefined flight boundaries.
6	206126011	206126025		Develop a drone simulation that surveys a parking lot, detects occupied and vacant parking spaces, and generates a parking availability report
7	206126018	206126008	206126007	Develop a simulation in which a drone searches a predefined disaster zone for simulated human targets.
8	206126035	206126033	206126034	Simulate a drone delivering a medical package to a remote location while avoiding obstacles and handling a temporary route blockage
9	206126030	206126006	206126014	Simulate a drone surveying a solar farm and detecting panels with artificially introduced visual anomalies
10	206126010	206126024	206126029	Simulate a drone navigating through a room containing furniture and moving obstacles. Implement a collision-avoidance method
11	306426001	206126020	206126021	Simulate a drone surveying a public park or beach. Use object detection to identify litter in camera images and create a map of detected waste locations.
12	206126004	206126015		Simulate a drone to capture aerial images or video of public places, such as railway stations, stadiums, festivals, or campus events. Integrate a computer vision model to detect people, estimate crowd density, and identify regions with high crowd concentration.
13	406326003	406326004	411125053	Develop Autonomous navigation system for a drone and also add the feature return the drone to home when battery is low (provide alert)
	Instructions			

1. Submit the softcopy of the document containing Team details (Roll No., Name), Question no., Problem statement, Design, Tool Used, Environment / platform
2. Demo video (less than 2 minutes)
3. Set of procedures to be followed for execution with all files & code with read me file
4. Contribution of each member of the team
5. **Submission of softcopy : 10.12.2026 before 11 am**
6. **Submission details:**

   https://classroom.google.com/c/ODk3MDYwNzQ0MDgw?cjc=yqdffn6w

   Class code: yqdffn6w
7. **Show live demo on or before 07.12.2026**
8. Use any simulator/open software/ tool of your choice